

FBR / PRAL Scenarios — Sale Type & Tax Rate Reference

InvoiceSolution · Digital Invoicing · Generated from source (apps/fbr/scenarios.py + builder.py)

Source of truth: Scenario titles + builders → backend/apps/fbr/scenarios.py · Sale-type strings → backend/apps/fbr/sale_types.py (must match PRAL /pdi/v1/transtypecode verbatim) · UoM strings → backend/apps/fbr/builder.py (UOM_FBR_MAP) · Auto-selection → pick_scenario_id .

How scenarios work. In **sandbox**, PRAL assigns each taxpayer a set of scenario IDs; every invoice line's saleType must be compatible with the scenarioId sent, or PRAL rejects with 0204 ("Sale type not match with provided scenario"). In **production** there is no scenarioId — PRAL classifies each line by its saleType alone. The app picks the sandbox scenario automatically from the invoice's contents; the rate is set per line and must be one PRAL allows for that saleType .

The full table (SN001–SN028)

Scenario	Title	saleType (exact)	Tax rate	Notes
SN001	Standard Rate Goods → Registered Buyers	Goods at standard rate (default)	18%	B2B, input-tax-credit eligible
SN002	Standard Rate Goods → Unregistered Buyers	Goods at standard rate (default)	18%	Walk-in / B2C
SN003	Steel (Melted / Re-Rolled)	Steel melting and re-rolling	18%	Sector-specific
SN004	Steel Scrap by Ship Breakers	Ship breaking	18%	
SN005	Reduced Rate Goods (8th Schedule)	Goods at Reduced Rate	1%	8th Schedule
SN006	Exempt Goods (6th Schedule)	Exempt goods	Exempt	rate = literal "Exempt" , not "0%"
SN007	Zero-Rated Goods (5th Schedule)	Goods at zero-rate	0%	5th Schedule
SN008	3rd Schedule Goods	3rd Schedule Goods	18%	tax computed on retail/MRP
SN009	Purchase from Cotton Ginners	Cotton ginners	18%	
SN010	Telecom Services by Mobile Operators	Telecommunication services	17%	
SN011	Steel via Toll Manufacturing	Toll Manufacturing	18%	
SN012	Petroleum Products	Petroleum Products	1.43%	
SN013	Electricity to Retailers	Electricity Supply to Retailers	5%	
SN014	Gas to CNG Stations	Gas to CNG stations	18%	
SN015	Mobile Phones	Mobile Phones	18%	
SN016	Processing / Conversion of Goods	Processing/Conversion of Goods	5%	
SN017	Goods (FED in ST Mode)	Goods (FED in ST Mode)	8%	
SN018	Services (FED in ST Mode)	Services (FED in ST Mode)	8%	
SN019	Services (ICT Ordinance)	Services	16% (also 0/Exempt/5/15/17%)	NOT 18% — see Services section
SN020	Electric Vehicles	Electric Vehicle	1%	SRO: 6th Schd Table III
SN021	Cement / Concrete Block	Cement /Concrete Block	Rs.3 per unit	fixed per-unit
SN022	Potassium Chlorate	Potassium Chlorate	18% + Rs.60/kg	mixed

Scenario	Title	saleType (exact)	Tax rate	Notes
SN023	CNG Sales	CNG Sales	Rs.200/unit	SRO: 581(1)/2024
SN024	Goods per SRO 297(I)/2023	Goods as per SRO.297(I)/2023	25%	saleType uses a pipe , not letter I
SN025	Drugs (8th Schedule Serial 81)	Non-Adjustable Supplies	0%	SRO: EIGHTH SCHEDULE Table 1
SN026	Standard Rate → End Consumers (Retail)	Goods at standard rate (default)	18%	retail variant of SN002
SN027	3rd Schedule → End Consumers (Retail)	3rd Schedule Goods	18%	tax on retail; retail variant of SN008
SN028	Reduced Rate → End Consumers (Retail)	Goods at Reduced Rate	1%	SRO: EIGHTH SCHEDULE Table 1

Services (SN019) — for service providers

Services HS codes live under **chapter 98** (98xx.xxxx), e.g. 9819.1300 (Commission Agents). A services line has **three** requirements that differ from goods, all confirmed against the live PRAL sandbox validator:

Field	Goods (wrong for services)	Services (correct)
saleType	Goods at standard rate (default)	Services
uoM	Numbers, pieces, units	Others
scenarioId (sandbox)	SN002	SN019
rate	18%	16% (valid: 0% / Exempt / 5% / 15% / 16% / 17%)

Getting any one wrong yields, respectively: 0099 (UoM not allowed for HS), 0204 (sale type / scenario mismatch), 0046 (rate not valid for sale type).

The app now handles the first three automatically. When a chapter-98 HS code is used, it sends `saleType = Services`, `uoM = Others`, and picks `SN019`. **The operator must still choose a services-valid rate** — pick **"Services 16%"** (or 15%) from the tax-rate dropdown; **18% is rejected for services**.

Rates that are NOT a simple percentage

- **SN021** — Rs.3 per unit (fixed per unit)
- **SN023** — Rs.200/unit (fixed per unit)
- **SN022** — 18% + Rs.60/kg (ad-valorem + specific)
- **SN012** — 1.43% (petroleum)

3rd-Schedule math (SN008 / SN027)

For 3rd-Schedule (retail-price-fixed) lines, sales tax is charged **ON the MRP** for the given rate, not extracted tax-inclusive:

```
fixedNotifiedValueOrRetailPrice = retail_price × qty
salesTaxApplicable               = retail_total × rate / 100
valueSalesExcludingST           = retail_total      # full MRP (PRAL rejects 0/empty)
totalValues                      = retail_total + salesTaxApplicable
```

Auto-selection (pick_scenario_id)

In **sandbox**, the app derives the `scenarioId` from the whole invoice (production sends none). All-lines-of-a-kind → that scenario; otherwise the standard fallback:

Invoice content	Scenario picked
All exempt lines (Exempt goods)	SN006 (if assigned)
All zero-rate lines (Goods at zero-rate)	SN007 (if assigned)
All 3rd-Schedule lines (fixedNotifiedValueOrRetailPrice > 0)	SN027 (walk-in, if assigned) else SN008
All reduced-rate lines (8th Schedule)	SN028 (walk-in, if assigned) else SN005
All services lines (HS ch. 98 / saleType = Services)	SN019
Registered buyer, standard goods	SN001
Unregistered buyer, standard goods	SN026 (retail, if assigned) else SN002

Retail end-consumer scenarios (SN026/27/28) are only used for **walk-in (unregistered)** buyers **and** only when the tenant is actually assigned that code in IRIS. Otherwise the app falls back to the non-retail equivalent.

Gotchas

- **rate is a string** like "18%", not a number. Exempt sends the literal "Exempt" .
- **uoM is a verbose enum string** — "KG" not "Kilograms", "Numbers, pieces, units" for each, "Others" for services.
- **saleType must match /pdi/v1/transtypecode verbatim**, including the pipe character in SN024.
- **scenarioId is sandbox-only** — omitted in production.
- The sandbox test rates in build_snXXX are the canonical rate for each scenario; a real invoice's rate is set per line and must be one PRAL allows (query /pdi/v1/SaleTypeToRate for the exact list).